

Payment Integration System architecture

The available architectures in the market for project buildings are

1. Monolith architecture
2. Service oriented architecture(SOA)
3. Microservice based architecture
4. Modular monolith architecture(Spring modulith framework)

We are going to design the architecture using microservice based architecture

Our system will mainly have 3 microservices

1. Payment validation service
2. Payment processing service
3. Stripe provider service

The functionalities of the microservices

1. Payment validation service

- All incoming client requests are first routed to this service
- Responsible for validating the request before further processing
- Performs two main types of validations:
 - **Field Validation** (e.g., required fields, data formats)
 - **Business Validation** (e.g., payment rules, limits, eligibility)
- If the request is valid, it is forwarded for further processing
- If invalid, the request is rejected with an appropriate response

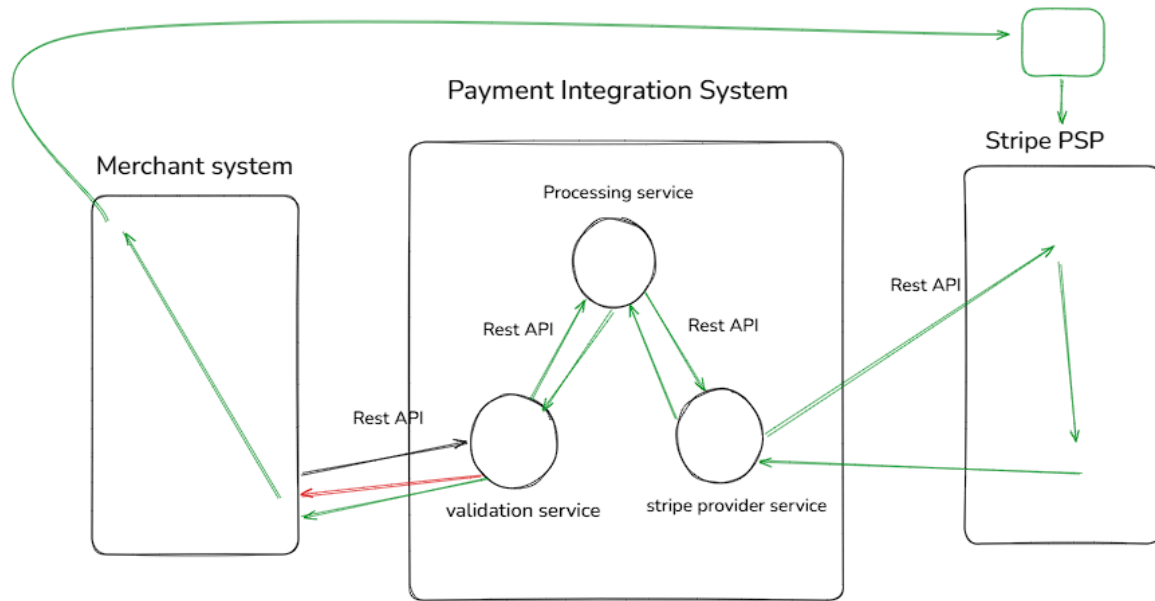
2. Payment processing service

- Responsible for overall payment management
- Identifies the appropriate payment provider (e.g., Stripe, PayPal, Trustly, etc.)
- Routes and calls the selected provider service based on the integration type
- Acts as the central layer coordinating different payment integrations

3. Stripe provider service

- Responsible for integrating with the Stripe payment system

- All communication with Stripe is handled within this microservice
- Implements Create Session API integration
- Handles webhook processing (incoming requests from Stripe are received and processed here)



Communication between the services are handled by using RestAPI

Note:

These 3 microservice communicate to each other via Rest API

Without some web API communication one service cannot invoke another